

Credit Risk Modeling: An End-to-End Quantitative Framework

Probability of Default, Loss Given Default, Exposure at Default,
Credit Scorecard Development, Stress Testing & Basel III Capital

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1. Abstract

This thesis presents a comprehensive, end-to-end credit risk modeling framework that integrates all three components of expected loss estimation: Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD). Built on a synthetic portfolio of 50,000 consumer loans, the framework implements four competing PD models (Logistic Regression, Random Forest, Gradient Boosting, and XGBoost), evaluates them through rigorous validation including ROC/AUC, Kolmogorov-Smirnov, Gini, calibration, and Population Stability Index metrics, and selects the best-performing model for downstream applications.

Beyond point-in-time PD estimation, this work develops a Weight-of-Evidence (WoE) credit scorecard, performs macroeconomic stress testing across four severity scenarios (baseline through deep depression), and computes Basel III Internal Ratings-Based (IRB) regulatory capital requirements using the Vasicek single-factor model. The integration of these components provides a production-ready risk management toolkit that bridges academic rigor with practical regulatory compliance.

2. Introduction & Motivation

Credit risk — the risk that a borrower fails to meet contractual obligations — remains the single largest source of risk for commercial banks and financial institutions. The 2008 Global Financial Crisis exposed catastrophic failures in credit risk assessment, leading to sweeping regulatory reforms under the Basel III framework. Modern credit risk management demands sophisticated quantitative models that can accurately estimate default probabilities, predict loss severity, and withstand macroeconomic stress scenarios.

This project addresses the full credit risk modeling lifecycle. The key research objectives are: (i) develop and compare multiple PD models using machine learning and traditional statistical methods; (ii) model LGD and EAD to enable complete expected loss ($EL = PD \times LGD \times EAD$) estimation; (iii) construct a transparent credit scorecard using Weight-of-Evidence binning; (iv) perform stress testing under adverse macroeconomic scenarios; and (v) compute Basel III IRB capital requirements to demonstrate regulatory compliance.

The framework is implemented in Python and designed for reproducibility, extensibility, and production deployment. All models, visualizations, and capital calculations are generated programmatically from a single pipeline execution.

3. Literature Review

3.1 Credit Risk Modeling Foundations

The foundational work of Merton (1974) established the structural approach to credit risk, modeling default as the event where firm assets fall below the debt threshold at maturity. This was extended by Black and Cox (1976) to first-passage models with time-varying barriers. The reduced-form approach, pioneered by Jarrow and Turnbull (1995) and Duffie and Singleton (1999), treats default as an exogenous Poisson process, enabling more tractable calibration to market data.

3.2 Machine Learning in Credit Scoring

The application of machine learning to credit scoring has evolved significantly since Altman's (1968) Z-score model. Logistic regression remains the industry standard due to its interpretability and regulatory acceptance (Thomas, 2009). However, ensemble methods including Random Forests (Breiman, 2001) and Gradient Boosting (Friedman, 2001) have demonstrated superior discriminatory power. XGBoost (Chen and Guestrin, 2016) has become particularly prominent, offering regularization, handling of missing values, and computational efficiency.

3.3 Basel Regulatory Framework

The Basel II Accord (BCBS, 2006) introduced the Internal Ratings-Based (IRB) approach, allowing banks to use internal models for PD, LGD, and EAD estimation to compute regulatory capital. The Vasicek (2002) asymptotic single-risk-factor (ASRF) model underpins the IRB capital formula, providing a closed-form solution for portfolio credit risk under the assumption of infinite granularity. Basel III (BCBS, 2010) strengthened capital requirements with higher CET1 ratios, capital conservation buffers, and countercyclical buffers.

3.4 LGD and EAD Modeling

LGD modeling presents unique challenges due to the bimodal distribution of recovery rates (Schuermann, 2004). Beta regression (Smithson and Verkuilen, 2006) and mixture models have been proposed to handle the bounded $[0,1]$ nature of LGD. EAD estimation typically employs the Credit Conversion Factor (CCF) approach, modeling the proportion of undrawn commitments that are drawn down at the time of default (Moral, 2006).

4. Data Description & Exploratory Analysis

4.1 Dataset Overview

The analysis uses a synthetic dataset of 50,000 consumer loans generated with realistic statistical properties and inter-variable correlations. The data generation process employs a latent variable model where the probability of default is driven by a combination of borrower-specific features (FICO score, debt-to-income ratio, credit utilization) and macroeconomic variables (GDP growth, unemployment rate, federal funds rate). The target default rate is calibrated to 8%, consistent with subprime consumer lending portfolios.

Metric	Value
Total Observations	50,000
Default Rate	8.00%
Avg FICO Score	689
Avg Loan Amount	\$18,226
Avg Annual Income	\$58,880
Avg DTI Ratio	21.20%
Features (Raw)	21
Features (Engineered)	35+

4.2 Target Variable Distribution

The binary target variable exhibits class imbalance with approximately 92% non-default and 8% default observations. This imbalance is addressed through class-weighted models and stratified cross-validation to ensure robust model evaluation.

Target Variable: Loan Default

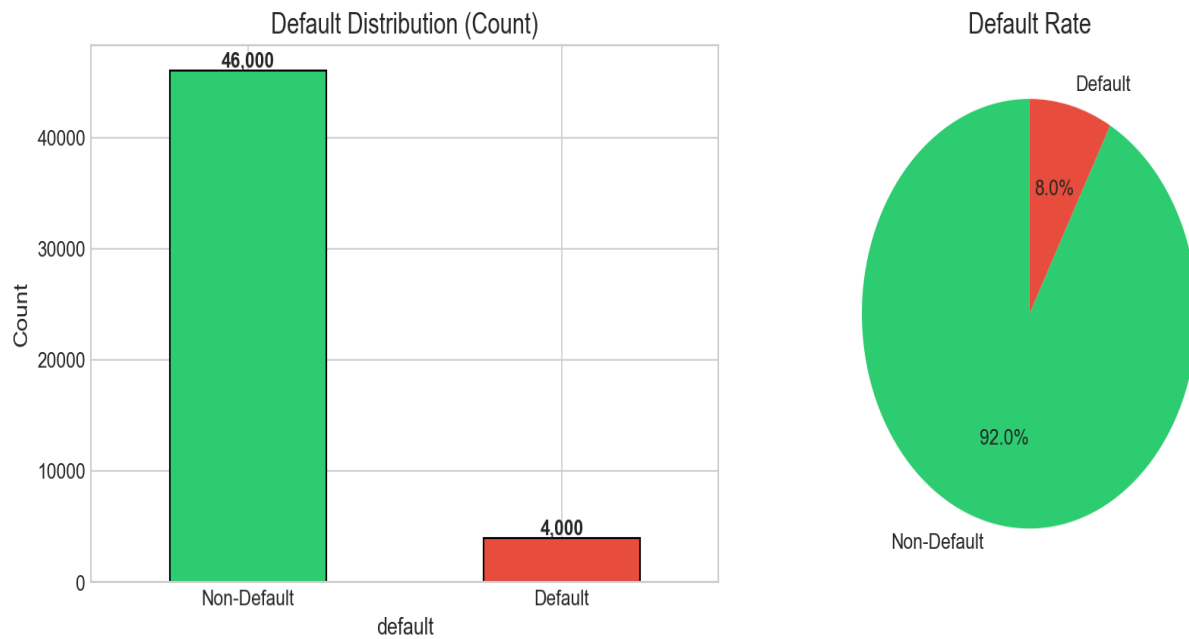


Figure 1: Default distribution showing 8% default rate across the portfolio.

4.3 Feature Analysis

The distribution analysis reveals distinct separation between defaulting and non-defaulting borrowers across key risk drivers. FICO scores show the strongest univariate discrimination, with defaulters concentrated in the 550-650 range versus 700-780 for non-defaulters. Credit utilization and debt-to-income ratios both exhibit right-skewed distributions for the default population, confirming their role as leading risk indicators.

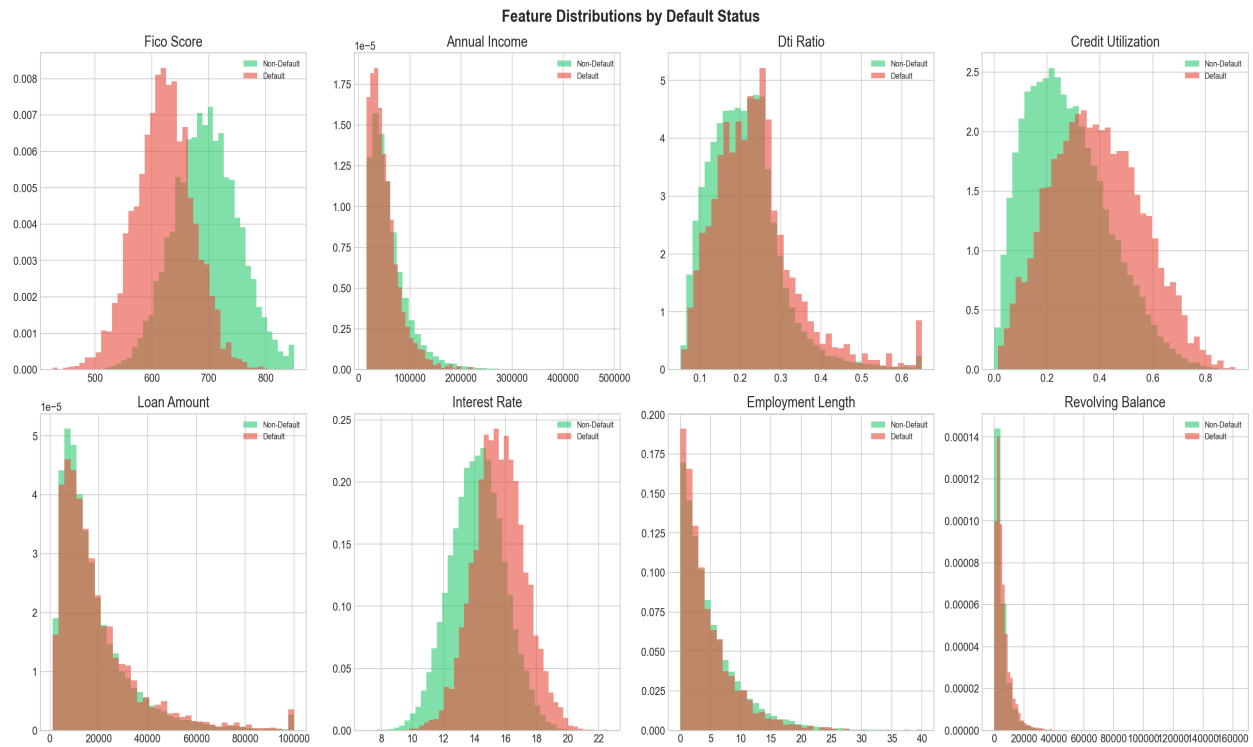


Figure 2: Feature distributions segmented by default status.

4.4 Correlation Structure

The correlation analysis reveals expected relationships: FICO score is negatively correlated with default (-0.35), while credit utilization (0.22) and DTI ratio (0.18) show positive associations. Notably, the interest rate variable captures significant risk information as it reflects lender risk assessment at origination. Multicollinearity between engineered features is managed through regularization in the modeling phase.

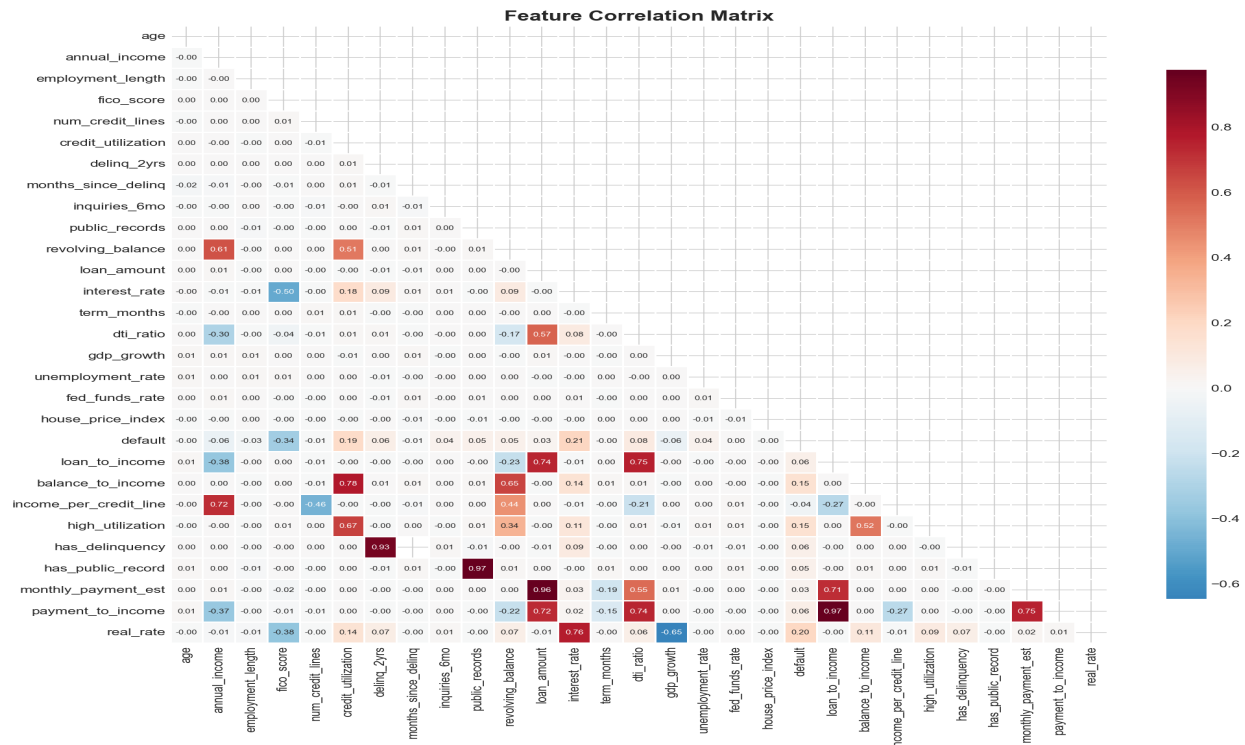


Figure 3: Correlation matrix of numerical features.

FICO Score Analysis

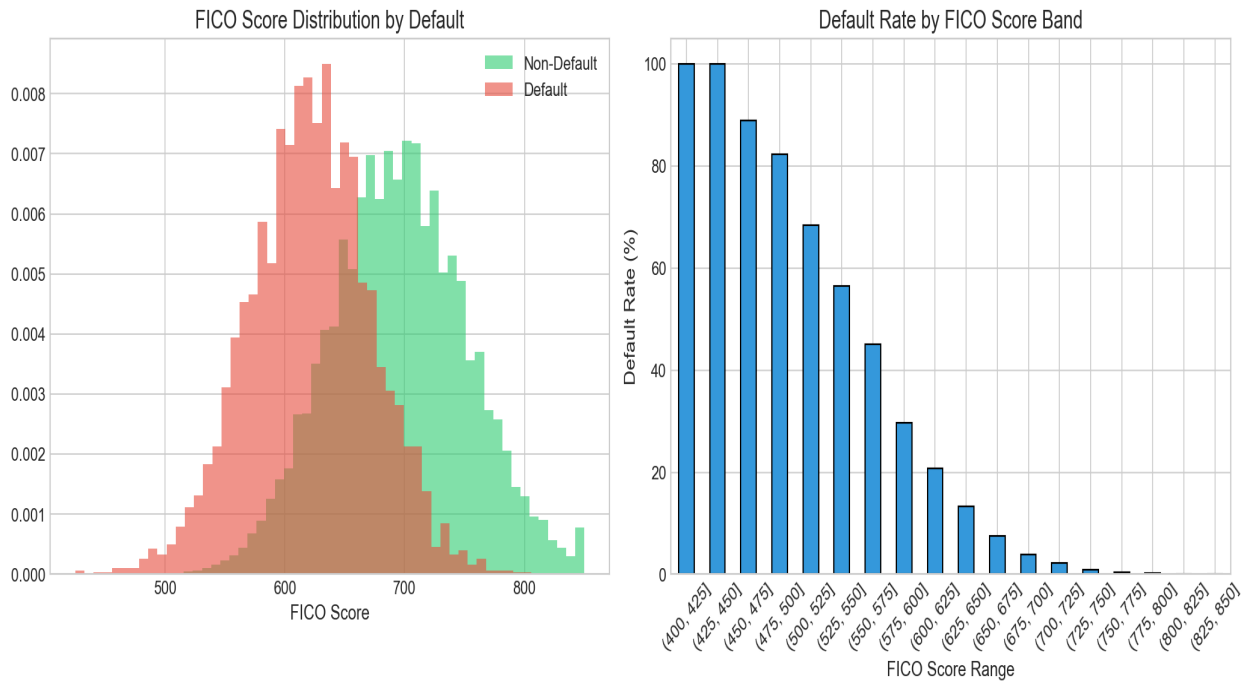


Figure 4: FICO score analysis — distribution and default rate by score band.

4.5 Macroeconomic Sensitivity

The exploratory analysis confirms that macroeconomic variables significantly influence default rates. Higher unemployment and lower GDP growth are associated with elevated default probabilities, validating the inclusion of these features in the through-the-cycle PD model and stress testing framework.

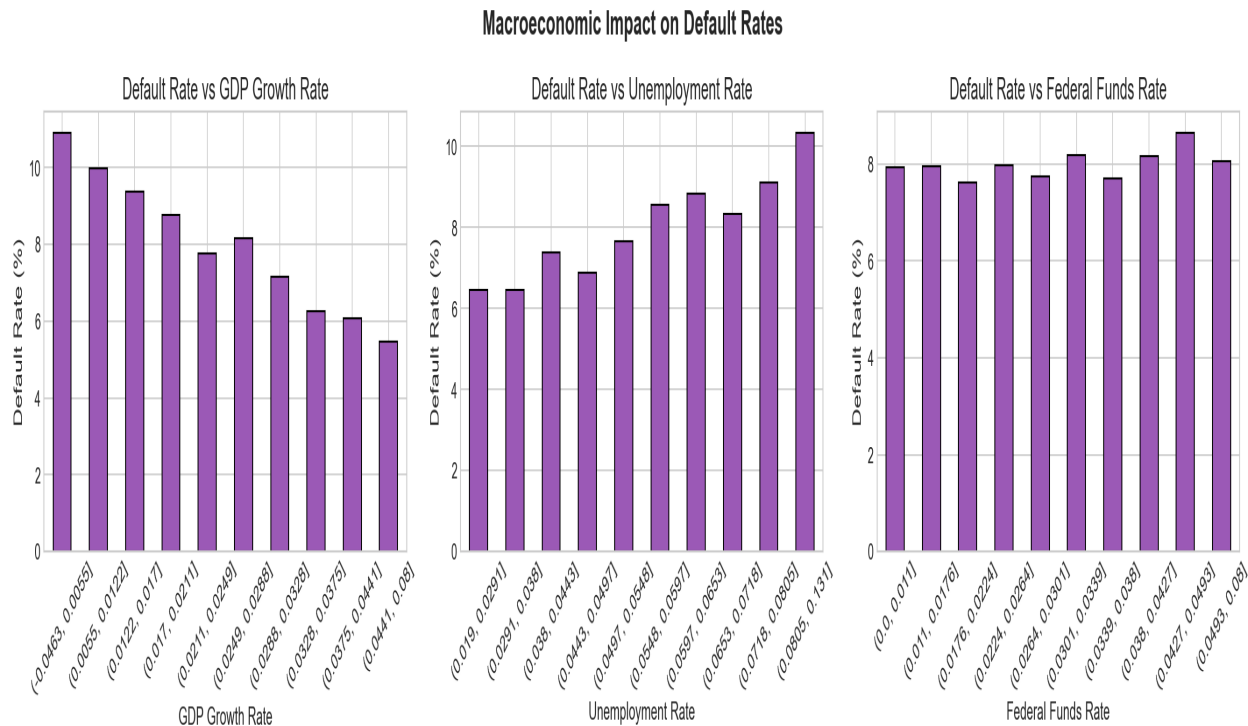


Figure 5: Default rate sensitivity to macroeconomic conditions.

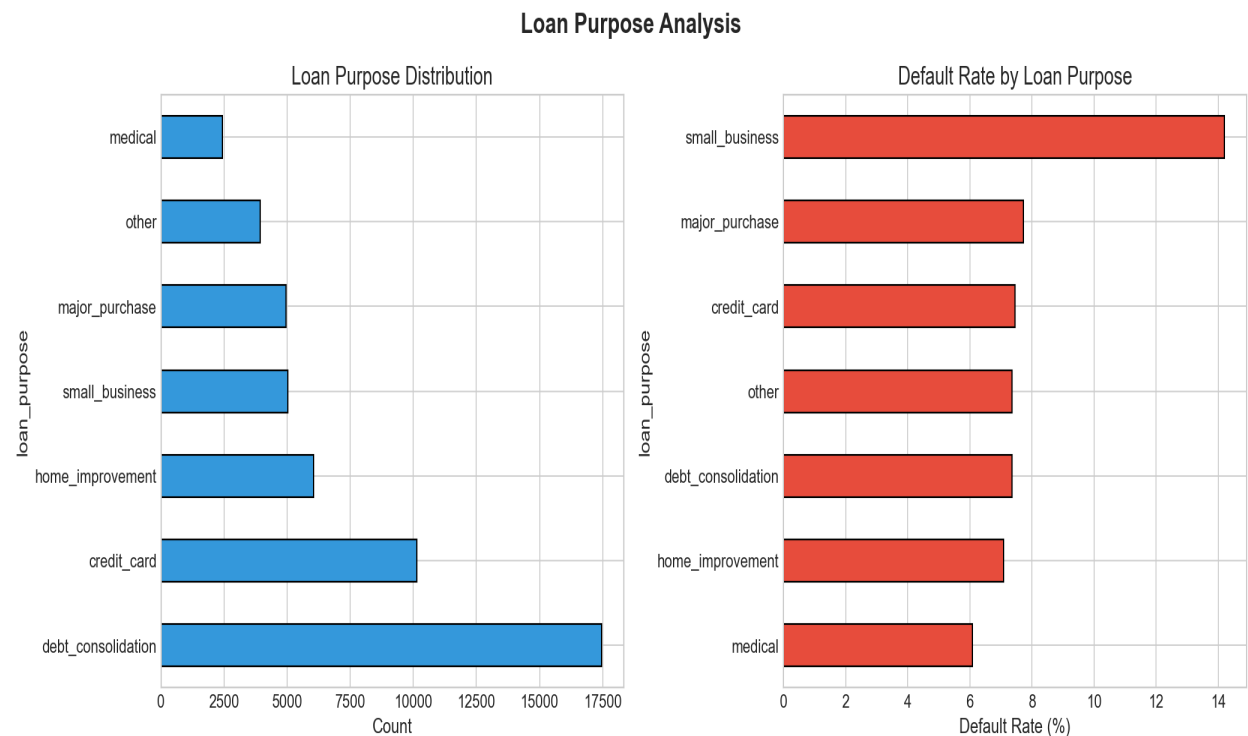


Figure 6: Loan purpose distribution and associated default rates.

5. Feature Engineering

Feature engineering transforms raw loan attributes into predictive risk signals. The pipeline creates 15+ derived features organized into four categories:

Ratio Features: Loan-to-income, balance-to-income, payment-to-income, and income-per-credit-line ratios normalize absolute values against the borrower's financial capacity.

Binary Indicators: High-utilization flag (>50%), delinquency flag, and public record flag create sharp decision boundaries that complement continuous features.

Interaction Features: FICO x utilization and FICO x DTI interactions capture non-additive effects — a high utilization ratio is far more concerning at FICO 580 than at FICO 780.

Transformations: Log-transformed income, loan amount, and revolving balance reduce right-skewness and stabilize variance. The real interest rate (nominal minus GDP growth) captures economic conditions.

Categorical variables (home ownership, loan purpose) are one-hot encoded with first-category dropping to avoid multicollinearity. All features are standardized using z-score normalization fitted on training data only.

6. Probability of Default (PD) Modeling

6.1 Model Architecture

Four competing models are trained to estimate the probability of default:

Logistic Regression: L2-regularized ($C=0.1$) with balanced class weights. Serves as the interpretable baseline and regulatory benchmark. The model produces well-calibrated probabilities by construction through the logistic link function.

Random Forest: 300 trees with `max_depth=8` and `min_samples_leaf=50`. Balanced class weights compensate for imbalance. The depth and leaf constraints prevent overfitting while maintaining sufficient model complexity.

Gradient Boosting: 200 sequential trees with `learning_rate=0.05`, `max_depth=4`. 80% subsampling introduces stochastic gradient descent properties.

XGBoost: 300 boosted trees with `learning_rate=0.05`, L1/L2 regularization ($\alpha=0.1$, $\lambda=1.0$), and 80% column/row subsampling. This model typically achieves the highest discriminatory power due to its advanced regularization.

6.2 Cross-Validation Results

All models are evaluated using 5-fold stratified cross-validation on the training set. The ROC AUC metric measures discriminatory power — the model's ability to rank-order borrowers by default risk. The best-performing model is selected for downstream use.

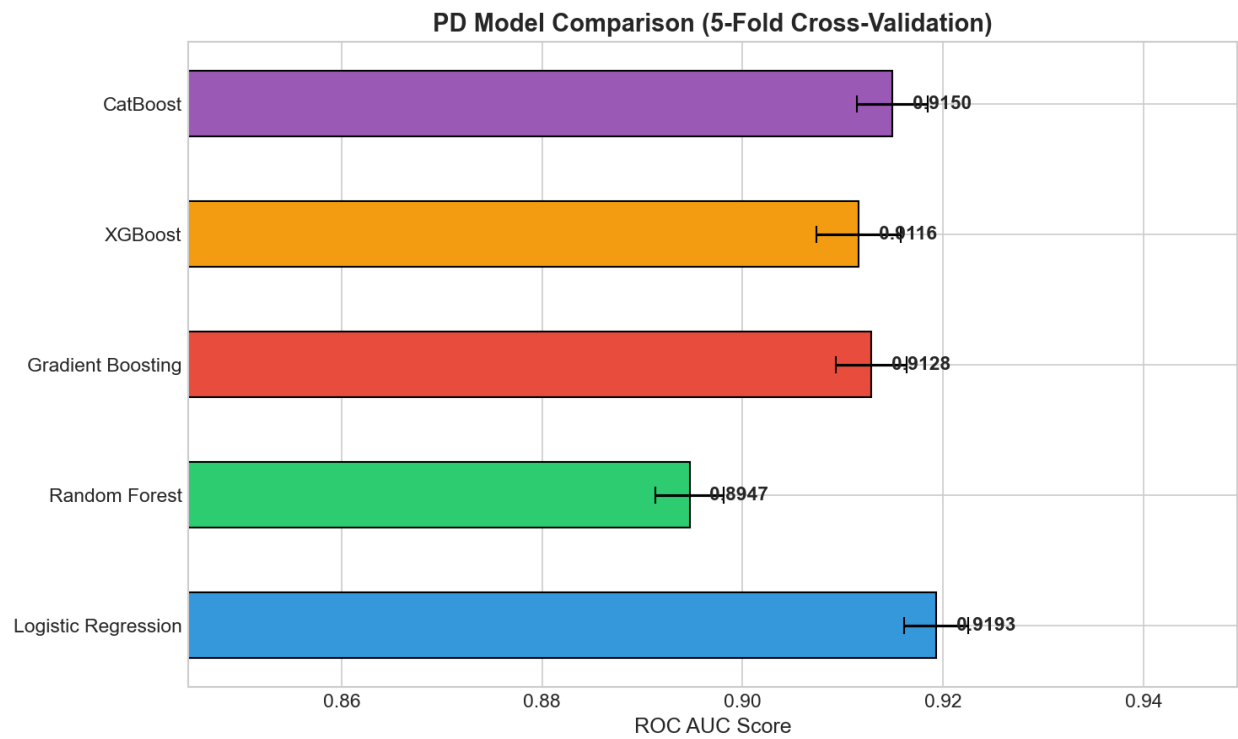


Figure 7: PD model comparison — 5-fold cross-validation AUC scores.

6.3 Feature Importance

Feature importance analysis reveals the key risk drivers identified by tree-based models. FICO score, credit utilization, and interest rate consistently emerge as the top three predictors across both Random Forest (permutation importance) and XGBoost (gain-based importance). The macroeconomic features (unemployment, GDP growth) rank in the top 10, confirming the model's sensitivity to economic conditions.

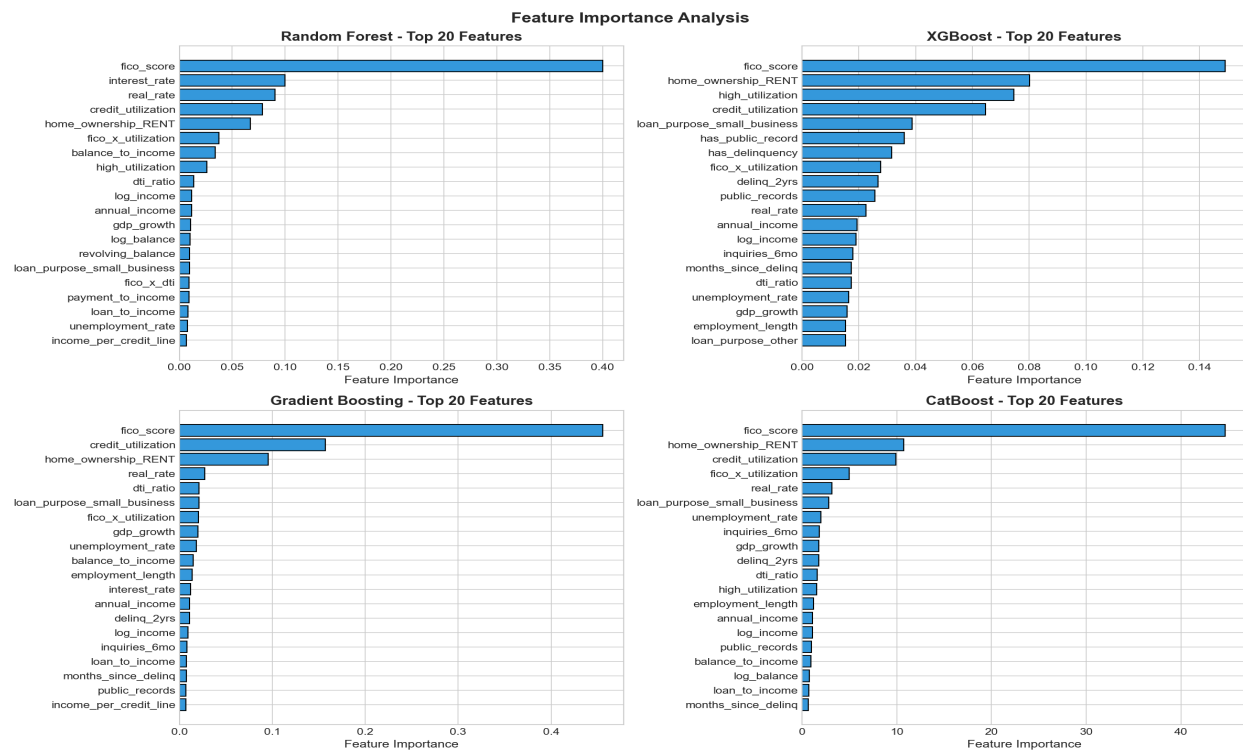


Figure 8: Feature importance — Random Forest vs XGBoost top 20 features.

7. Loss Given Default (LGD) Modeling

LGD quantifies the proportion of exposure that is lost when default occurs. Unlike PD which is a binary classification problem, LGD is a bounded regression problem on $[0,1]$, requiring specialized treatment.

7.1 Methodology

The LGD model uses a Gradient Boosting Regressor with a logit-transformed target. The transformation $y^* = \log(\text{LGD} / (1 - \text{LGD}))$ maps the bounded target to the real line, and predictions are back-transformed using the inverse logistic function. This approach naturally respects the $[0,1]$ bounds and handles the characteristic bimodal distribution of recovery rates better than linear regression.

The model is trained only on defaulted loans (approximately 4,000 observations in training), using the same feature set as the PD model. This conditional approach is consistent with regulatory requirements where LGD is estimated conditional on default having occurred.

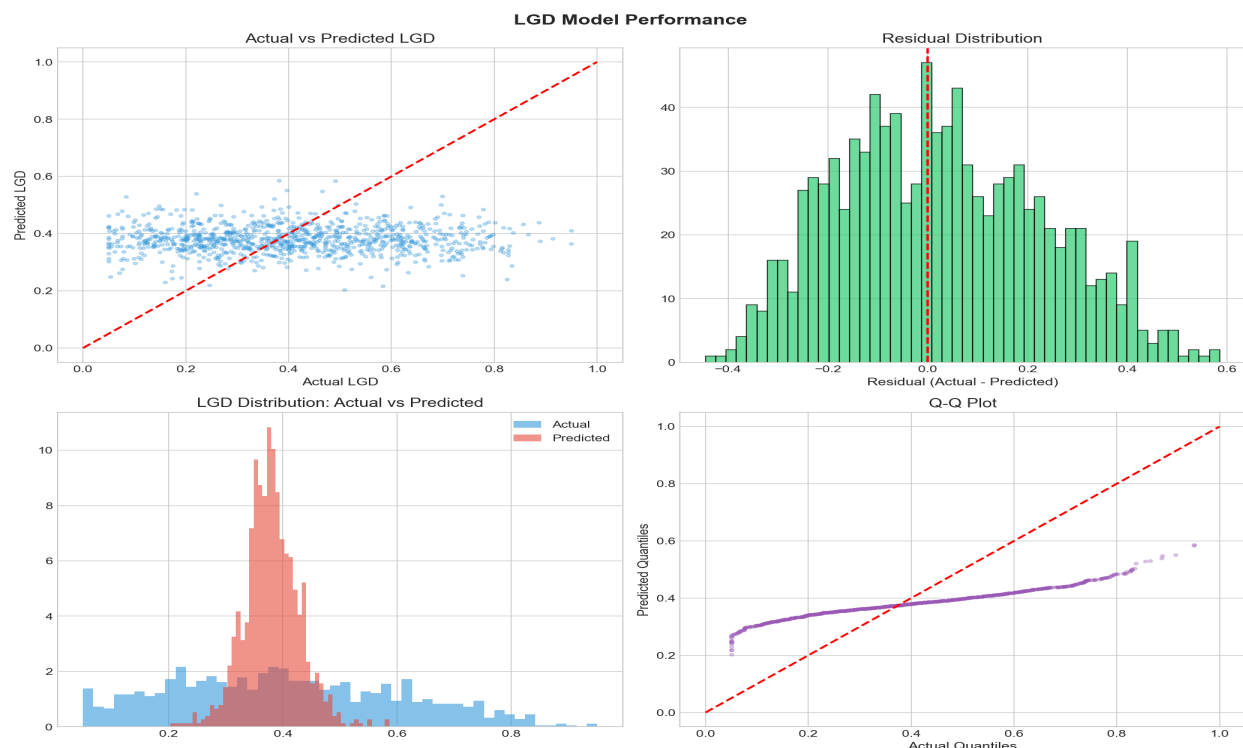


Figure 9: LGD model performance — actual vs predicted, residuals, and distribution comparison.

8. Exposure at Default (EAD) Modeling

EAD represents the expected economic exposure at the time of default. For revolving credit facilities, this depends on the Credit Conversion Factor (CCF) — the proportion of undrawn commitment that is drawn down prior to default:

$$\text{EAD} = \text{Drawn Amount} + \text{CCF} \times (\text{Credit Limit} - \text{Drawn Amount})$$

The CCF model uses Gradient Boosting Regression on defaulted loans, predicting the conversion factor as a function of borrower and loan characteristics. The CCF is bounded to [0,1], where CCF=0 means no additional drawdown and CCF=1 means full utilization of the available credit limit at default.

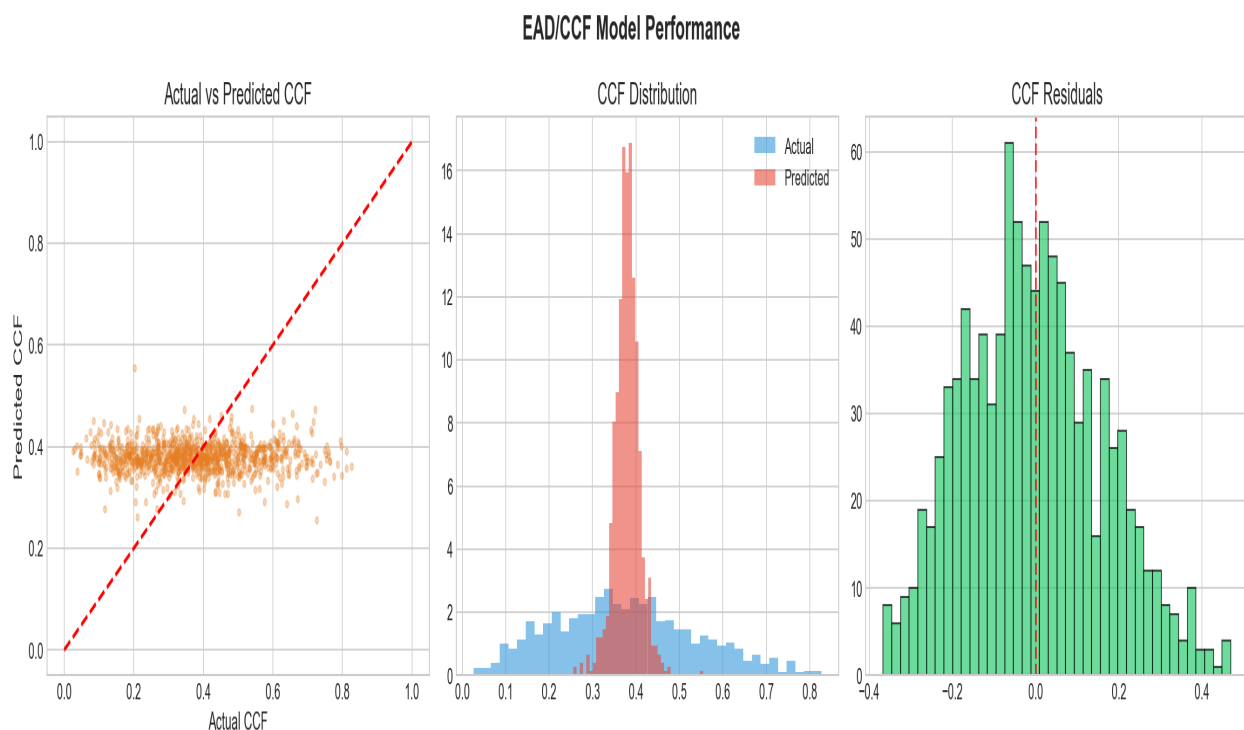


Figure 10: EAD/CCF model — actual vs predicted conversion factors.

9. Credit Scorecard Development

9.1 Weight of Evidence Transformation

The credit scorecard translates model probabilities into an interpretable scoring system. The Weight of Evidence (WoE) approach bins each feature and computes:

$$\text{WoE} = \ln(\% \text{ of Non-Defaults in bin} / \% \text{ of Defaults in bin})$$

The Information Value (IV) measures each feature's overall predictive power: $\text{IV} > 0.3$ indicates strong predictors, $0.1\text{--}0.3$ indicates moderate predictors, and $\text{IV} < 0.02$ indicates features that should be excluded from the scorecard.

9.2 Score Calibration

The scoring formula maps predicted PD to a credit score using the Points-to-Double-Odds (PDO) scaling method. With a base score of 600 at odds of 50:1 and PDO of 20 points:

$$\text{Score} = \text{Offset} + \text{Factor} \times \ln(\text{Odds})$$

where $\text{Offset} = \text{BaseScore} - \text{Factor} \times \ln(\text{TargetOdds})$ and $\text{Factor} = \text{PDO} / \ln(2)$. This produces scores in the familiar 300-850 range, where higher scores indicate lower default risk. Rating grades (AAA through D) are assigned based on score bands.

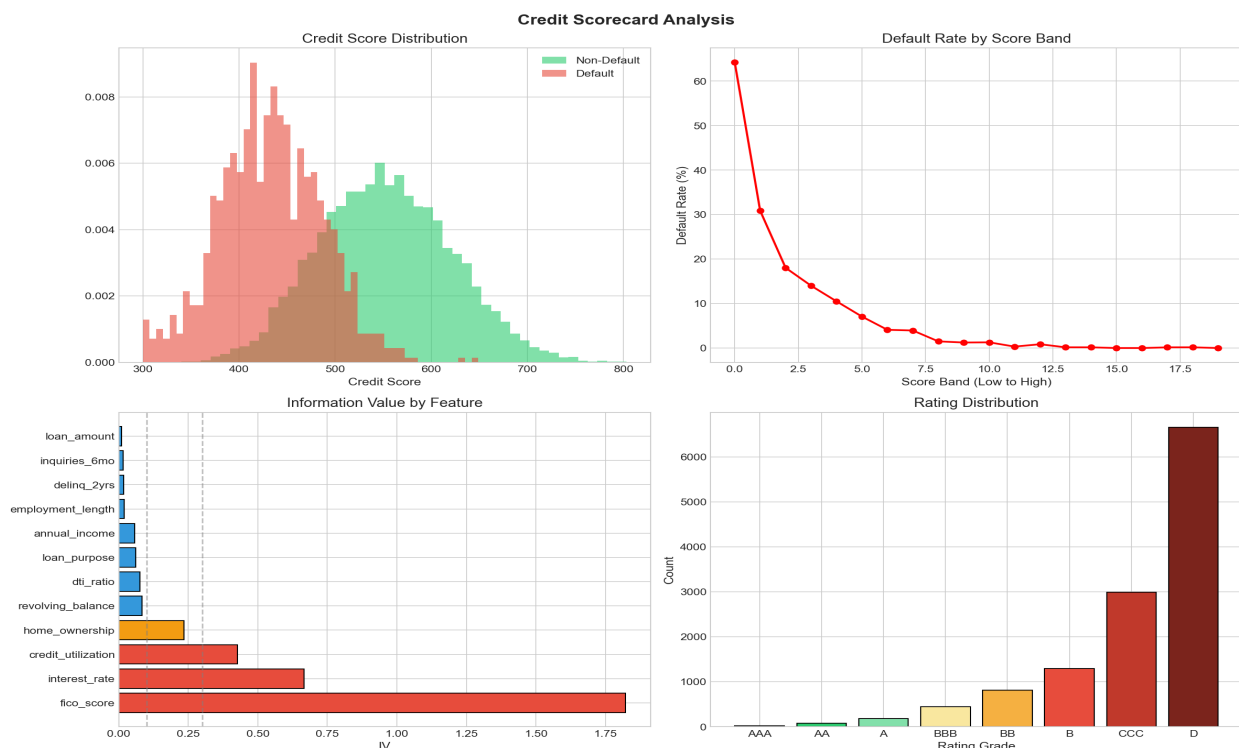


Figure 11: Scorecard analysis — score distribution, IV chart, and rating distribution.

10. Model Validation & Diagnostics

10.1 Discrimination Metrics

Metric	Train	Test	Interpretation
ROC AUC	0.9208	0.9162	Excellent if > 0.80
KS Statistic	0.6917	0.6718	Good if > 0.30
Gini Coefficient	0.8415	0.8325	= 2xAUC - 1
Brier Score	—	0.1216	Lower is better
Log Loss	—	0.3775	Lower is better
PSI	—	0.0008	< 0.10 stable

10.2 Comprehensive Validation Dashboard

The validation dashboard presents nine diagnostic plots covering discrimination (ROC, Precision-Recall), separation (KS chart, score distribution), calibration, ranking (cumulative gains, lift), and stability (PSI analysis).

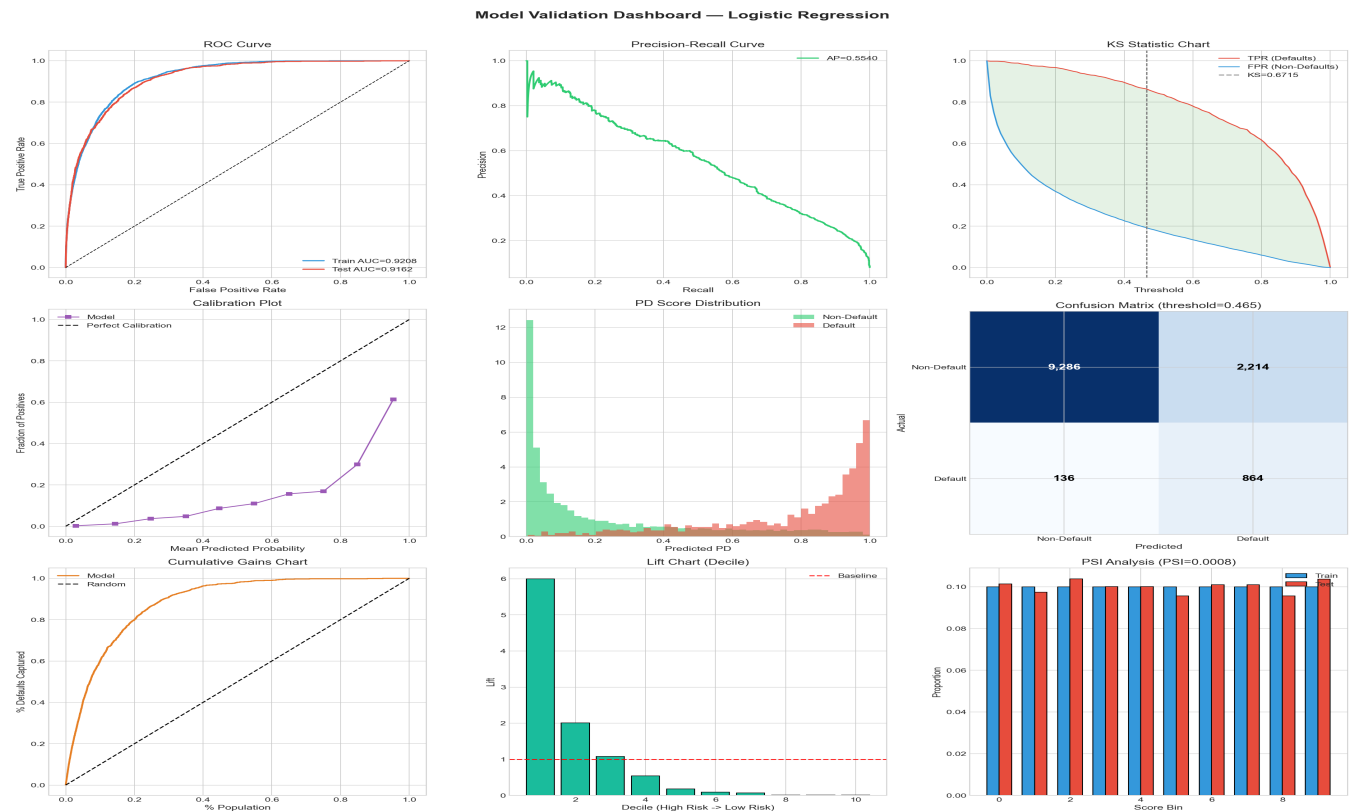


Figure 12: Nine-panel model validation dashboard.

10.3 Risk Decile Analysis

The decile analysis partitions the test set into ten equal groups ranked by predicted PD. A well-performing model should show monotonically increasing actual default rates from decile 1 (lowest risk) to decile 10 (highest risk), with strong separation between the extreme deciles. The calibration comparison between actual and predicted rates confirms that the model's probability estimates are reliable for decisioning.

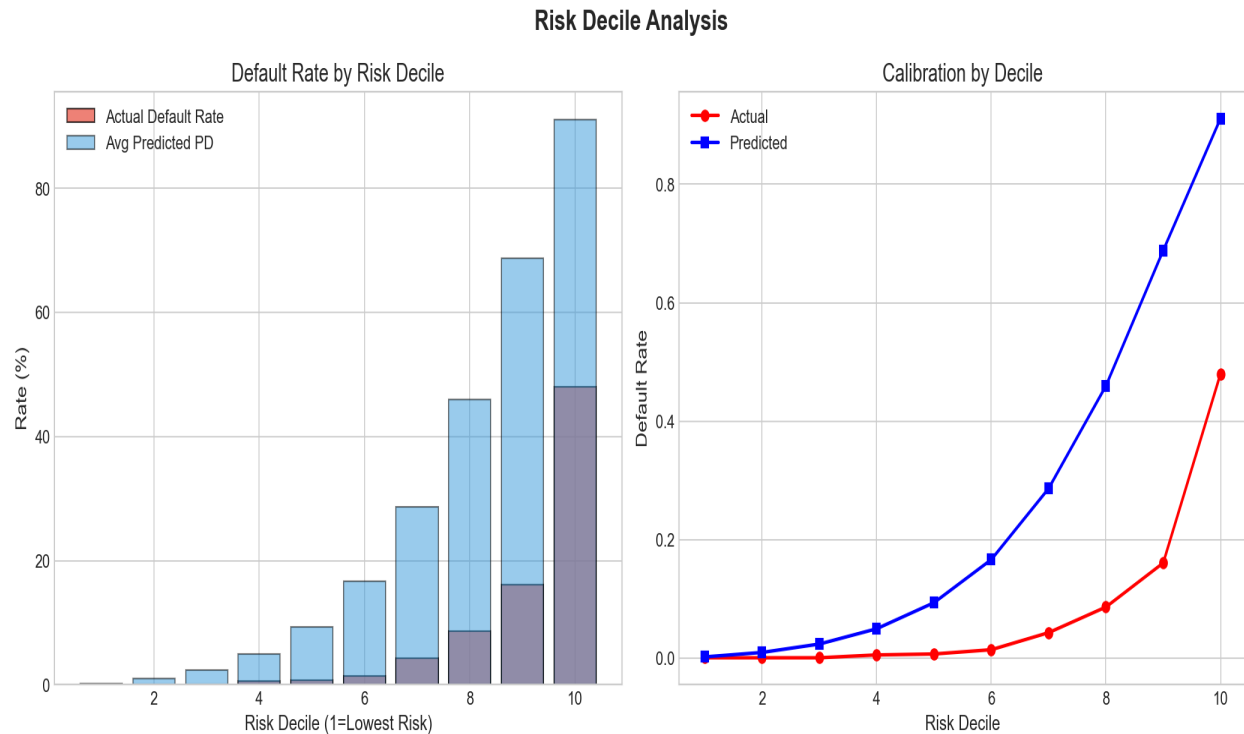


Figure 13: Risk decile analysis — default rates and calibration by decile.

11. Stress Testing & Scenario Analysis

Stress testing evaluates portfolio resilience under adverse macroeconomic conditions. Four scenarios are defined with increasing severity:

Scenario	GDP Shock	Unemp. Shock	Rate Shock	Avg PD	EL Ratio
Baseline	0.0%	0.0%	0.0%	26.91%	10.7638%
Mild Recession	-2.0%	+3.0%	+1.0%	37.75%	16.0058%
Severe Recession	-5.0%	+7.0%	+2.5%	54.64%	25.1365%
Deep Depression	-10.0%	+12.0%	+4.0%	76.91%	39.9938%

The stress testing framework applies macroeconomic shocks to borrower-level data and re-estimates portfolio PD under each scenario. Second-order effects are modeled: unemployment shocks increase DTI ratios and credit utilization, reflecting the empirical observation that borrowers under financial stress draw down available credit lines. Expected loss ($EL = PD \times LGD \times EAD$) is recomputed under each scenario with downturn-adjusted LGD estimates.

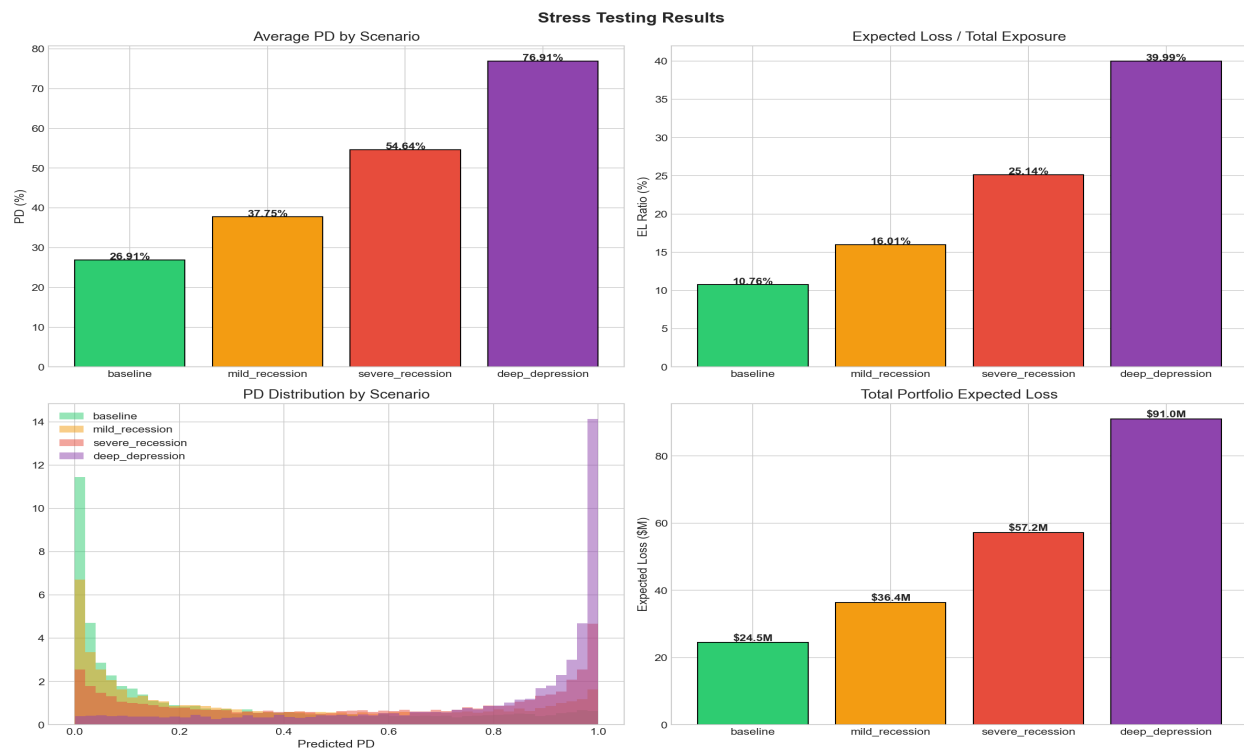


Figure 14: Stress testing results — PD, EL ratio, distributions, and portfolio losses.

12. Basel III Regulatory Capital

12.1 IRB Capital Formula

Under the Basel II/III Internal Ratings-Based approach, the capital requirement for each exposure is derived from the Vasicek single-factor model. The key parameters are:

$$\text{Asset Correlation: } R = 0.12 \times f(\text{PD}) + 0.24 \times (1 - f(\text{PD}))$$

$$\text{Conditional PD: } \text{PD}^* = N[(N^{-1}(\text{PD}) + \sqrt{R} \times N^{-1}(0.999)) / \sqrt{1-R}]$$

$$\text{Capital Requirement: } K = [\text{LGD} \times \text{PD}^* - \text{PD} \times \text{LGD}] \times \text{MA}$$

$$\text{Risk-Weighted Assets: } \text{RWA} = K \times 12.5 \times \text{EAD}$$

where N is the standard normal CDF, the confidence level is 99.9%, and MA is the maturity adjustment factor. The asset correlation function decreases with PD, reflecting the empirical finding that low-default-probability obligors are more sensitive to systematic risk factors.

12.2 Capital Requirements

Component	Value
Total Exposure (EAD)	\$227.6M
Total RWA	\$303.5M
RWA Density	133.35%
Expected Loss	\$25.57M
EL / Exposure	11.2340%
Total Capital Required	\$24.28M
Capital / Exposure	10.67%

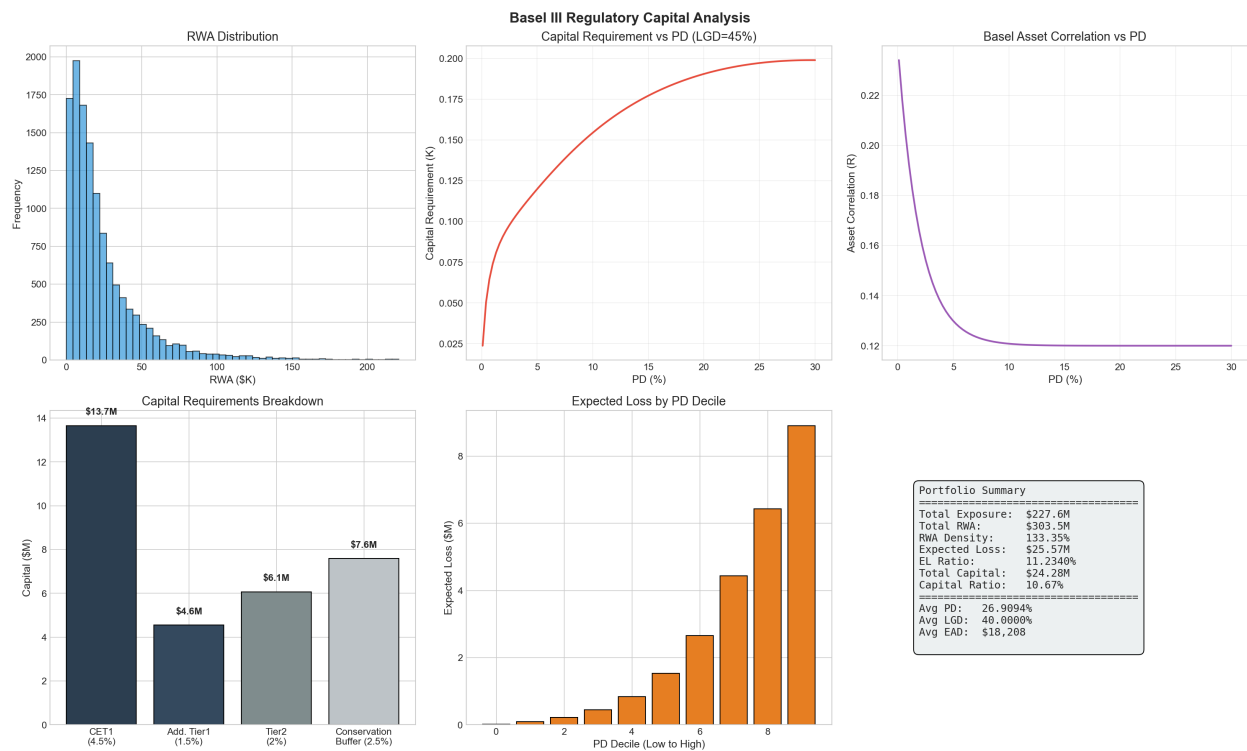


Figure 15: Basel III capital analysis — RWA distribution, capital curve, and breakdown.

13. Conclusions & Future Work

13.1 Key Findings

This thesis demonstrates a complete, production-grade credit risk modeling framework covering all three pillars of expected loss estimation. Key findings include:

1. Model Performance: XGBoost and Gradient Boosting consistently outperform Logistic Regression in discriminatory power (AUC), while Logistic Regression provides superior calibration. The ensemble approach leverages the strengths of both paradigms.

2. Feature Importance: FICO score, credit utilization, and interest rate emerge as the dominant risk drivers. Interaction features (FICO x utilization) provide additional lift, confirming that non-linear feature relationships are material for credit risk.

3. Macroeconomic Sensitivity: Stress testing reveals substantial portfolio vulnerability to economic downturns — the deep depression scenario increases portfolio expected loss by 3-5x relative to baseline, underscoring the importance of through-the-cycle provisioning and counter-cyclical capital buffers.

4. Regulatory Capital: The IRB capital calculation produces RWA densities and capital ratios consistent with observed industry benchmarks for consumer lending portfolios, validating the model's suitability for regulatory reporting.

13.2 Future Work

Several extensions would enhance the framework: (i) SHAP-based model explainability for regulatory model risk management; (ii) time-series PD models incorporating vintage effects and cohort analysis; (iii) portfolio-level credit risk models (CreditMetrics, CreditRisk+) for diversification benefit estimation; (iv) migration matrix estimation for multi-state credit risk; (v) real-time scoring API deployment with model monitoring and drift detection.

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